Solving Recurrence Relations and the Master Theorem

CMSC 27200: Theory of Algorithms

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We describe two ways of solving recurrence relations. The first way is to expand out the recurrence relation. The second way is to take an educated guess, check it, and adjust accordingly.

1 Expanding out recurrence relations

One way to solve recurrence relations is to iteratively substitute the recurrence relation into itself. This method is direct and intuitive, but it may not work as well for more complicated recurrence relations.

Example 1.1. *Consider the recurrence relation $T(n) = 2T(\frac{n}{2}) + n$ for mergesort. Repeatedly substituting this recurrence relation into itself, we have that*

1. $T(n) = 2T(\frac{n}{2}) + n$
2. $T(n) = 4T(\frac{n}{4}) + 2\frac{n}{2} + n$
3. $T(n) = 8T(\frac{n}{8}) + 4\frac{n}{4} + 2\frac{n}{2} + n$
4. $T(n) = 2^k T(\frac{n}{2^k}) + kn$

Now we take $k = log_2(n)$ so that $\frac{n}{2^k} = 1$. This gives us $T(n) = nT(1) + nlog_2(n)$. If $T(1) = 1$ then $T(n) = n + nlog_2(n)$. Thus, mergesort takes $O(nlogn)$ time.

Example 1.2. *Consider the recurrence relation $T(n) = T(\frac{n}{2}) + 1$ for binary search. Repeatedly substituting this recurrence relation into itself, we have that*

1. $T(n) = T(\frac{n}{2}) + 1$
2. $T(n) = T(\frac{n}{4}) + 1 + 1$
3. $T(n) = T(\frac{n}{8}) + 1 + 1 + 1$
4. $T(n) = T(\frac{n}{2^k}) + k$

Now we take $k = log_2(n)$ so that $\frac{n}{2^k} = 1$. This gives us $T(n) = T(1) + log_2(n)$. If $T(1) = 1$ then $T(n) = 1 + log_2(n)$. Thus, binary search takes $O(logn)$ time.

Example 1.3. *Consider the recurrence relation* $T(n) = 2T(\frac{n}{4}) + 10$*. Repeatedly substituting this recurrence relation into itself, we have that*

1. $T(n) = 2T(\frac{n}{4}) + 10$
2. $T(n) = 4T(\frac{n}{16}) + 2 * 10 + 10$
3. $T(n) = 8T(\frac{n}{64}) + 4 * 10 + 2 * 10 + 10$
4. $T(n) = 2^k T(\frac{n}{4^k}) + 10 \left(\sum_{j=0}^{k-1} 2^j\right)$

Now we take $k = log_4(n)$ *so that* $\frac{n}{4^k} = 1$*. Since* $2^{log_4(n)} = \sqrt{n}$ *and* $10\left(\sum_{j=0}^{k-1} 2^j\right) = 10(2^k - 1) = 10(\sqrt{n} - 1)$*, this gives us that*

$$T(n) = \sqrt{n}T(1) + 10(\sqrt{n} - 1) = (T(1) + 10)\sqrt{n} - 10$$

If $T(1) = 1$ *then* $T(n) = 11\sqrt{n} - 10$

2 The homogeneous and inhomogeneous parts of a recurrence relation

In order to describe how to take educated guesses for the solution to recurrence relations, we must first analyze the structre of the solutions we are looking for. Here we consider recurrence relations which are of the form $L(T, n) = g(n)$ where $L(T, n)$ is linear in T.

Definition 2.1. *We say that* $L(T, n)$ *is linear in* T *if for all* $T_1, T_2 : \mathbb{N} \to \mathbb{R}$ *and all* $a, b \in \mathbb{R}$

$$L(aT_1 + bT_2, n) = aL(T_1, n) + bL(T_2, n)$$

Example 2.2. $L(T, n) = T(n) - 2T(\frac{n}{2})$ *is linear in* T *as for all* T_1, T_2, a, b*,*

$$\begin{aligned} L(aT_1 + bT_2, n) &= aT_1(n) + bT_2(n) - 2aT_1(\frac{n}{2}) - 2bT_2(\frac{n}{2}) \\ &= aT_1(n) - 2aT_1(\frac{n}{2}) + bT_2(n) - 2bT_2(\frac{n}{2}) \\ &= aL(T_1, n) + bL(T_2, n) \end{aligned}$$

Definition 2.3. *Given a recurrence relation of the form* $L(T, n) = g(n)$ *where* $L(T, n)$ *is linear in* T*, we say that* $L(T, n)$ *is the homogeneous part of the recurrence relation and* $g(n)$ *is the inhomogeneous part of the recurrence relation.*

To find the general solution to a recurrence relation of the form $L(T, n) = g(n)$ where $L(T, n)$ is linear in T, we must do the following:

1. Find the vector space $\{T_0 : L(T_0, n) = 0\}$ of solutions to the homogeneous part of the recurrence relation.

2. Find a single solution T_1 to the entire recurrence relation $L(T_1, n) = g(n)$.

Proposition 2.4. *$T(n)$ is a solution to the recurrence $L(T, n) = g(n)$ if and only if we can write $T = T_1 + T_0$ where $L(T_0, n) = 0$*

Proof. If $T = T_1 + T_0$ then $L(T, n) = L(T_1, n) + L(T_0, n) = g(n) + 0 = g(n)$. Conversely, if $L(T, n) = g(n)$ then take $T_0 = T - T_1$. Now $T = T_1 + T_0$ and $L(T_0, n) = L(T, n) - L(T_1, n) = g(n) - g(n) = 0$ □

3 Educated guessing and checking

We now show how recurrence relations can be solved by taking educated guesses, checking them, and adusting accordingly.

For recurrence relations whose homogeneous part consists of terms of the form $T(cn)$ for some constant c, a good guess is $T_0(n) = n^p$ for some power p. The reason this is a good guess is because $T_0(cn)^p = c^p n^p$. The factors of n^p will cancel and then we just have to chose p so that the constants give us an equality.

To find a single solution to the entire recurrence relation, if $g(n)$ is a multiple of $n^{p'}$ then $T_1(n) = c' n^{p'}$ is often a good guess. The reason for this is that $T_1(cn) = c' c^{p'} n^{p'}$ so the $n^{p'}$ factors cancel and we can try and choose c' to obtain an equality. The main case when this fails is when $n^{p'}$ is a solution to the homogeneous part of the recurrence relation. When this is the case, it turns out that multiplying $g(n)$ by $c' log(n)$ works well.

Example 3.1. *For the recurrence relation $T(n) = 2T(\frac{n}{2}) + n$ for mergesort, we can take $L(T, n) = T(n) - 2T(\frac{n}{2})$ and $g(n) = n$.*

The homogeneous part of this recurrence relation is $L(T_0, n) = 0$ or equivalently $T_0(n) = 2T_0(\frac{n}{2})$. Plugging in $T_0(n) = n^p$ we obtain that $n^p = 2(\frac{n}{2})^p$ which is satisfied when $p = 1$. Thus, the vector space of solutions to the homogenous part is $\{T_0(n) = an : a \in \mathbb{R}\}$.

Here $g(n) = n$ is part of the solution to the homogeneous part so we instead try $T_1(n) = c' n log_2(n)$. Plugging this in we obtain that $c' n log_2(n) = 2c' \frac{n}{2}(log_2(n) - 1) + n$ which is true when $c' = 1$. Thus, the general solution to this recurrence relation is

$$T(n) = an + nlog_2(n)$$

If $T(1) = 1$ then solving for a we obtain that $a = 1$ and $T(n) = nlog_2(n) + n$

Example 3.2. *For the recurrence relation $T(n) = T(\frac{n}{2}) + 1$ for binary search, we can take $L(T, n) = T(n) - T(\frac{n}{2})$ and $g(n) = 1$.*

The homogeneous part of this recurrence relation is $L(T_0, n) = 0$ or equivalently $T_0(n) = T_0(\frac{n}{2})$. Plugging in $T_0(n) = n^p$ we obtain that $n^p = (\frac{n}{2})^p$ which is satisfied when $p = 0$. Thus, the vector space of solutions to the homogenous part is $\{T_0(n) = a : a \in \mathbb{R}\}$.

Here $g(n) = 1$ is part of the solution to the homogeneous part so we instead try $T_1(n) = c' log_2(n)$. Plugging this in we obtain that $c' log_2(n) = c'(log_2(n) - 1) + 1$ which is true when $c' = 1$. Thus, the general solution to this recurrence relation is

$$T(n) = a + log_2(n)$$

If $T(1) = 1$ then solving for a we obtain that $a = 1$ and $T(n) = log_2(n) + 1$

Example 3.3. *For the recurrence relation $T(n) = 2T(\frac{n}{4}) + 10$, we can take $L(T, n) = T(n) - 2T(\frac{n}{4})$ and $g(n) = 10$.*

The homogeneous part of this recurrence relation is $L(T_0, n) = 0$ or equivalently $T_0(n) = 2T_0(\frac{n}{4})$. Plugging in $T_0(n) = n^p$ we obtain that $n^p = 2(\frac{n}{4})^p$ which is satisfied when $p = \frac{1}{2}$. Thus, the vector space of solutions to the homogenous part is $\{T_0(n) = a\sqrt{n} : a \in \mathbb{R}\}$.

We now try $T_1(n) = c'$. Plugging this in we obtain that $c' = 2c' + 10$ which is true when $c' = -10$. Thus, the general solution to this recurrence relation is

$$T(n) = a\sqrt{n} - 10$$

If $T(1) = 1$ then the solution is $T(n) = 11\sqrt{n} - 10$

4 The Master Theorem

The Master theorem allows us to obtain the runtime of a recursive algorithm from the recurrence relation describing its runtime.

Theorem 4.1. *Let $T(n)$ be the worst-case runtime for an algorithm. If $T(n) = aT(\frac{n}{b}) + O(n^c)$ then letting $c_{crit} = log_b(a)$,*

1. *If $c < c_{crit}$ then $T(n)$ is $O(n^{c_{crit}})$*
2. *If $c > c_{crit}$ then $T(n)$ is $O(n^c)$*
3. *If $c = c_{crit}$ then $T(n)$ is $O(n^{c_{crit}} logn)$*

Proof sketch. We can solve the recurrence relation $T(n) = aT(\frac{n}{b}) + Cn^c$ and this will give an upper bound on $T(n)$. If $c \neq c_{crit}$, the solution to this recurrence relation is

$$T(n) = \left(T(1) - \frac{C}{1 - \frac{a}{b^c}}\right) n^{c_{crit}} + \frac{C}{1 - \frac{a}{b^c}} n^c$$

If $c < c_{crit}$ then this is $O(n^{c_{crit}})$. If $c > c_{crit}$ then this is $O(n^c)$

If $c = c_{crit}$ then the solution to this recurrence relation is

$$T(n) = T(1)n^{c_{crit}} + Cn^{c_{crit}} log_b(n)$$

which is $O(n^{c_{crit}} logn)$. $\square$

5 Advanced remarks

There are a few more things to note.

First, we previously said that we gave a general solution. Actually, this is not quite accurate. The reason is that our initial data only determines $T(n)$ for certain values of n. For example, if $T(n) = 2T(\frac{n}{2}) + n$ then $T(1)$ determines $T(2), T(4), T(8), etc.$ but does not determine $T(n')$ for other values of n'. It is quite possible that T behaves differently on $1, 2, 4, 8, \ldots$ than it does on $3, 6, 12, 24, \ldots$. Thus, strictly speaking we are only finding the general solution for T on a subset of the inputs n. For example, we found the general solution of $T(n) = 2T(\frac{n}{2}) + n$ on the inputs $n = 1, 2, 4, 8, \ldots$ and we found the general solution of $T(n) = 2T(\frac{n}{4}) + 10$ on the inputs $n = 1, 4, 16, 64, \ldots$. We sweep this detail under the rug for this course.

Second, we note that taking $T_0(n) = n^p$ is only a good guess when all of our linear terms have the form $T(cn)$ where c is a constant. If we instead only had terms of the form $T(n + c)$ where c is a constant then taking $T_0(n) = x^n$ is a good guess.

Example 5.1. *The Fibbonacci numbers obey the recurrence relation $F(n) = F(n-1)+F(n-2)$. Plugging in $F(n) = x^n$ we obtain that $x^n = x^{n-1}+x^{n-2}$ which is true if and only if $x^2-x-1 = 0$. The two solutions to this equation are $x = \frac{1\pm\sqrt{5}}{2}$. Thus, the general solution to this recurrence relation is $F(n) = c_1(\frac{1+\sqrt{5}}{2})^n + c_2(\frac{1-\sqrt{5}}{2})^n$.*

To find an explicit expression for the Fibonacci numbers, we can solve for c_1 and c_2 when $F(0) = 0$ and $F(1) = 1$. This gives us $c_1 = \frac{1}{\sqrt{5}}$ and $c_2 = -\frac{1}{\sqrt{5}}$ so the Fibonacci numbers are $F(n) = \frac{1}{\sqrt{5}}\left((\frac{1+\sqrt{5}}{2})^n - (\frac{1-\sqrt{5}}{2})^n\right)$

If we have different kinds of terms or a mixture of these kinds of terms, then a different guess may be needed or it may be impossible to give an explicit solution for the recurrence relation.

Third, if the recurrence relation has terms which depend in a non-linear way on T then the general solution no longer has the form $T = T_1 + T_0$ and it is generally difficult or impossible to find the general solution to such recurrence relations.

Example 5.2. *The recurrence relation $T(n) = T(n-1)^2$ is not linear in n because $T(n-1)$ is squared. One solution to this recurrence relation is $T(n) = 2^{2^n}$ as $(2^{2^{n-1}})^2 = 2^{2\cdot 2^{n-1}} = 2^{2^n}$. However, the general solution to this recurrence relation is not $c2^{2^n}$. Instead, the general solution to this recurrence relation is $T(n) = c^{2^n}$ as $(c^{2^{n-1}})^2 = c^{2\cdot 2^{n-1}} = c^{2^n}$*